

XMBA 200S: Data and Decisions

Chapter 21: The Simple Regression Model

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Course overview
Part 4: Building regression models

- Blocks 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best model you can build?
- This class: inference with the OLS regression
  - Next class: multivariate models

A bit more on the big picture…

Why regression models?

- Average effects of x on y often important for decision-making in a range of organizations

Pick a job, I'll give you a regression

Consultants:

zipcode_bank_deposits = b0 + b1*bank_competition



Pick a job, I'll give you a regression

HR: job_performance = b0 + b1*interview_score



Pick a job, I'll give you a regression

Marketing: reg revenues = b0 + b1*ad_spending

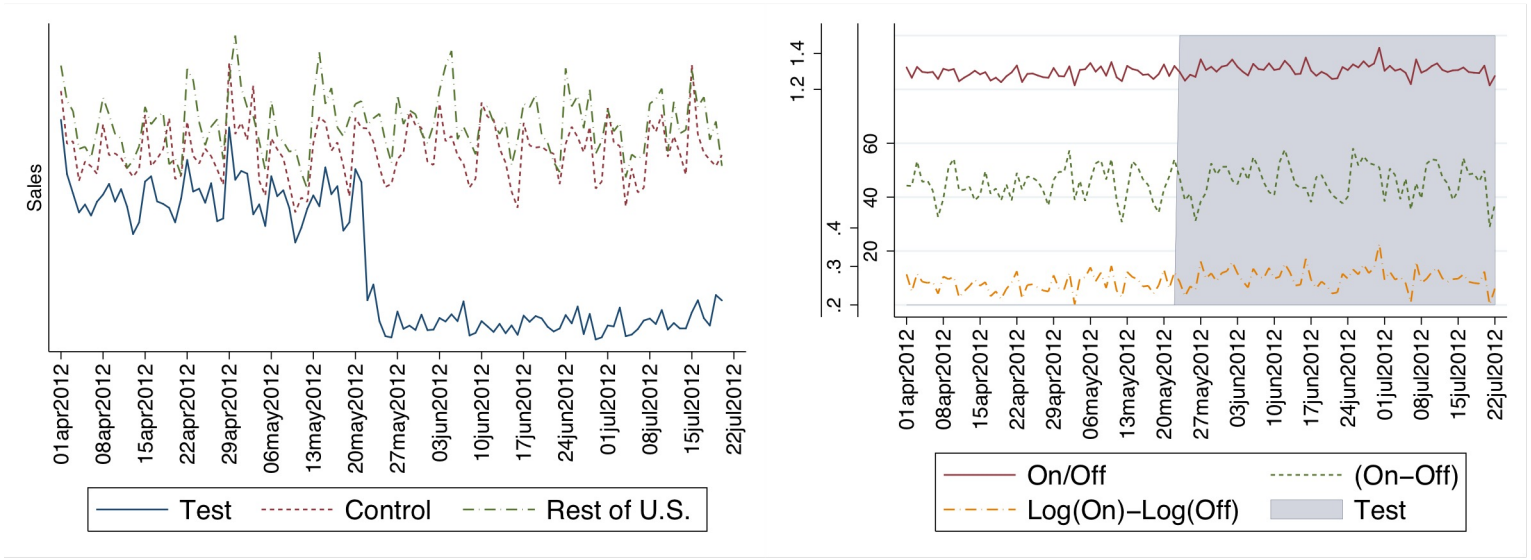


Figure 3: Non-Brand Keyword Region Test

(a) Attributed Sales by Region

(b) Differences in Total Sales

Pick a job, I'll give you a regression

Finance: share_price = b0 + b1*quarter_sales



Pick a job, I'll give you a regression

Education: test_scores = b0 + b1*math_curriculum



Pick a job, I'll give you a regression

- Politics: vote_share = b0 + b1*campaign_spending
- Recruiting: applicants = b0 + b1*unemployment_rate
- Firefighting: fire_damage = b0 + b1*wind_speed
- Non-Profits: donations = b0 + b1*interest_rates

Outline

- Setting up the simple regression model: theory
- Inference in regression models

Application: Locating a New UNIQLO Store



- You are working for UNIQLO
- You are put in charge of determining the location for a new retail store
  - You are considering a variety of different possible locations w/ different local traffic levels
- Find the relationship between local traffic levels (independent variable) & sales (dependent variable)

Location Choice Models





Caveat emptor: Location plays a crucial role in retail strategy in many industries. Real-world location models are more complex than the OLS framework presented in what follows & they incorporate spatial dimensions from which we will abstract.

Locating a New UNIQLO Store

You have sales data from a recent month obtained from n=80 UNIQLO stores. For each location you have sales and local traffic levels.

You regress sales on traffic levels.

The slope b_1 describes the average increase in sales for a given increase in traffic levels.

What would we predict for sales at a location with 50,000 cars driving by per day?



The Simple Regression Model

- Chapters 19 and 20 use regression to describe a sample.
- Today: Simple Regression Model (SRM), a model for the association in the population between an explanatory variable x and response y.
- Consider the data to be a sample from a population.
- Had we drawn a different sample, how different would the results be?

The Simple Regression Model (SRM)

Linear on Average

A simple regression model describes how the conditional mean of y depends on x.

$$\mu_{y|x} = \beta_0 + \beta_1 x$$

Equation describes our best guess for y given a particular x.

We don't know β_0 and β_1 but we can estimate them using data.

- β_0 and β_1 are population parameters that we will estimate using the regression statistics b_0 and b_1

The Simple Regression Model (SRM)

Deviations from the Mean

The model recognizes that individual values of response variable will differ from their conditional means,

$$y = \beta_0 + \beta_1 x + \varepsilon.$$

The deviations of responses around the conditional mean are called errors, denoted ε.

- Errors can be both positive and negative but have a zero average.

Sampling Distribution of our Regression Statistics: TLDR

Statistics are noisy… they have sample to sample variability

Regression estimates are statistics…

Nerds have figured out what that "noise" is…

The regression intercept b_0 (statistic!) and slope b_1 (statistic!) => t-distribution, centered on population parameters β_0 and β_1

Fortunately, nerds have also written software that internalizes this jargon, calculates standard errors for these statistics, etc…

Outline

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- Inference in regression models

The Simple Regression Model

Estimates vs. Parameters

Data		SRM
b_0	Intercept	β_0
b_1	Slope	β_1
$\hat{y}$	Prediction	$\mu_{y\|x}$
e	Residual	ε
s_e	Var(Resid)	σ_ε

Inference in Regression

The Standard Error of the Slope

Describes the sample-to-sample variability of b_1

The standard error of b_1 is,

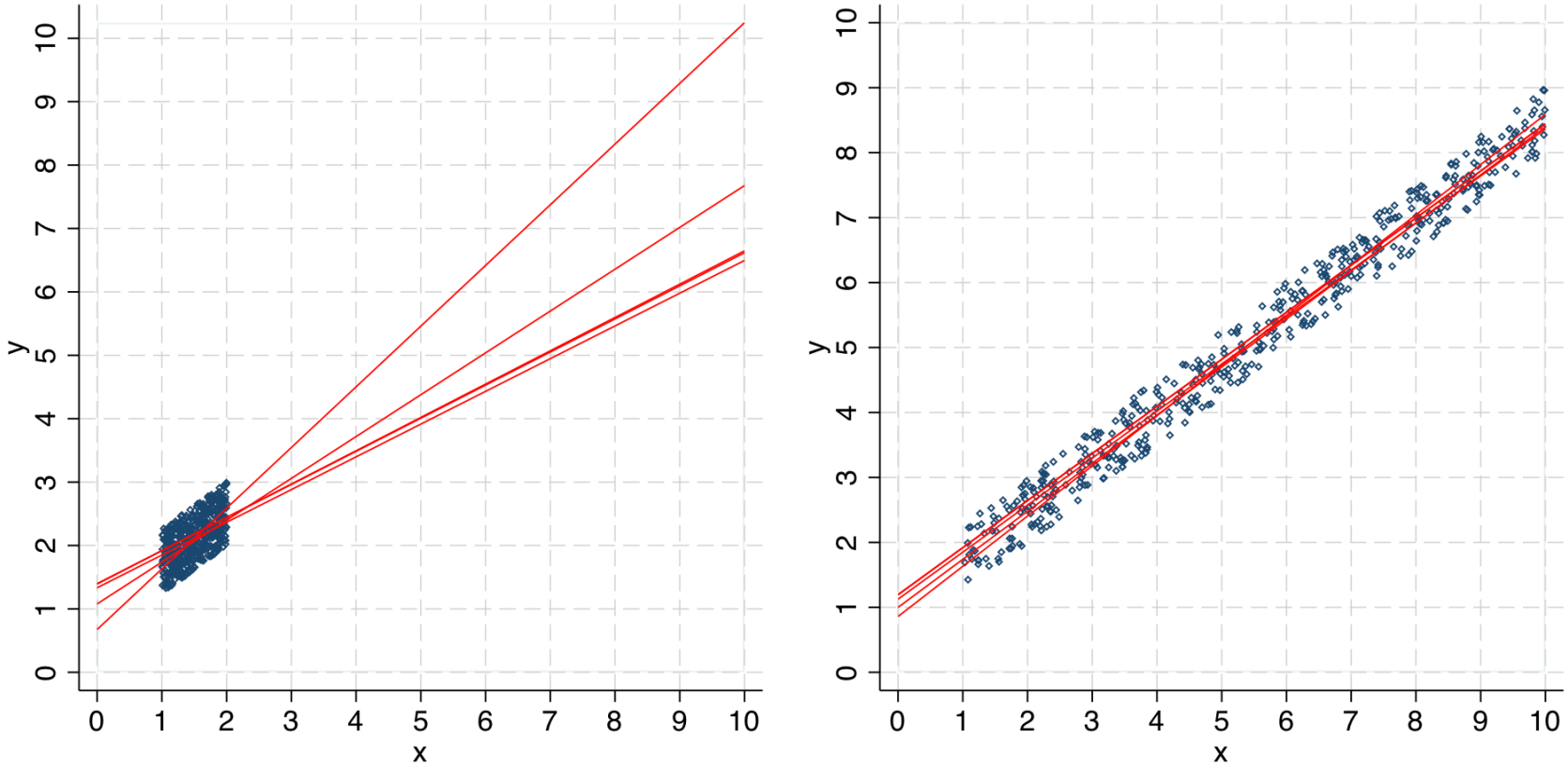
$$se(b_1) = \frac{s_e}{\sqrt{n-1}} * \frac{1}{s_x}$$

TLDR: The precision of our statistic b1 is related to:

1. sample size: n
2. how noisy the outcome data is: s_e
3. how noisy the explanatory data is: s_x

Inference in Regression

The Standard Error of the Slope



When s_x is larger, our the precision of our statistic b1 is higher. Why?

Inference in Regression

Estimated Standard Error of b_0

Feel free to squint or close your eyes

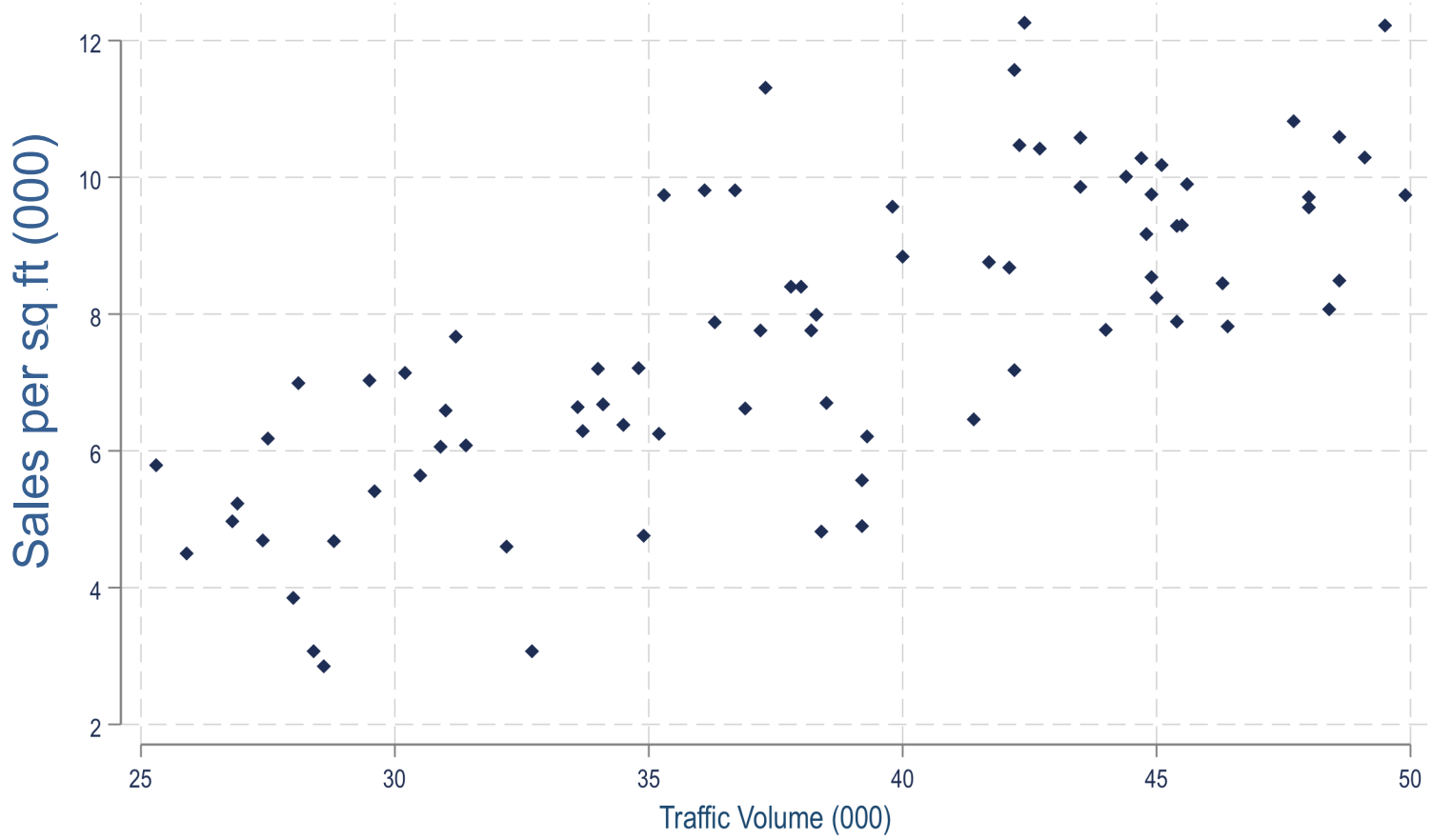
$$se(b_0) = s_e\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}} \approx \frac{s_e}{\sqrt{n}}\sqrt{1 + \frac{\bar{x}^2}{s_x^2}}$$

TLDR: The precision of our statistic b0 is related to:

- Decreases with sample size
- Increases with standard error of residual
- Increases with mean/variance tradeoff in x

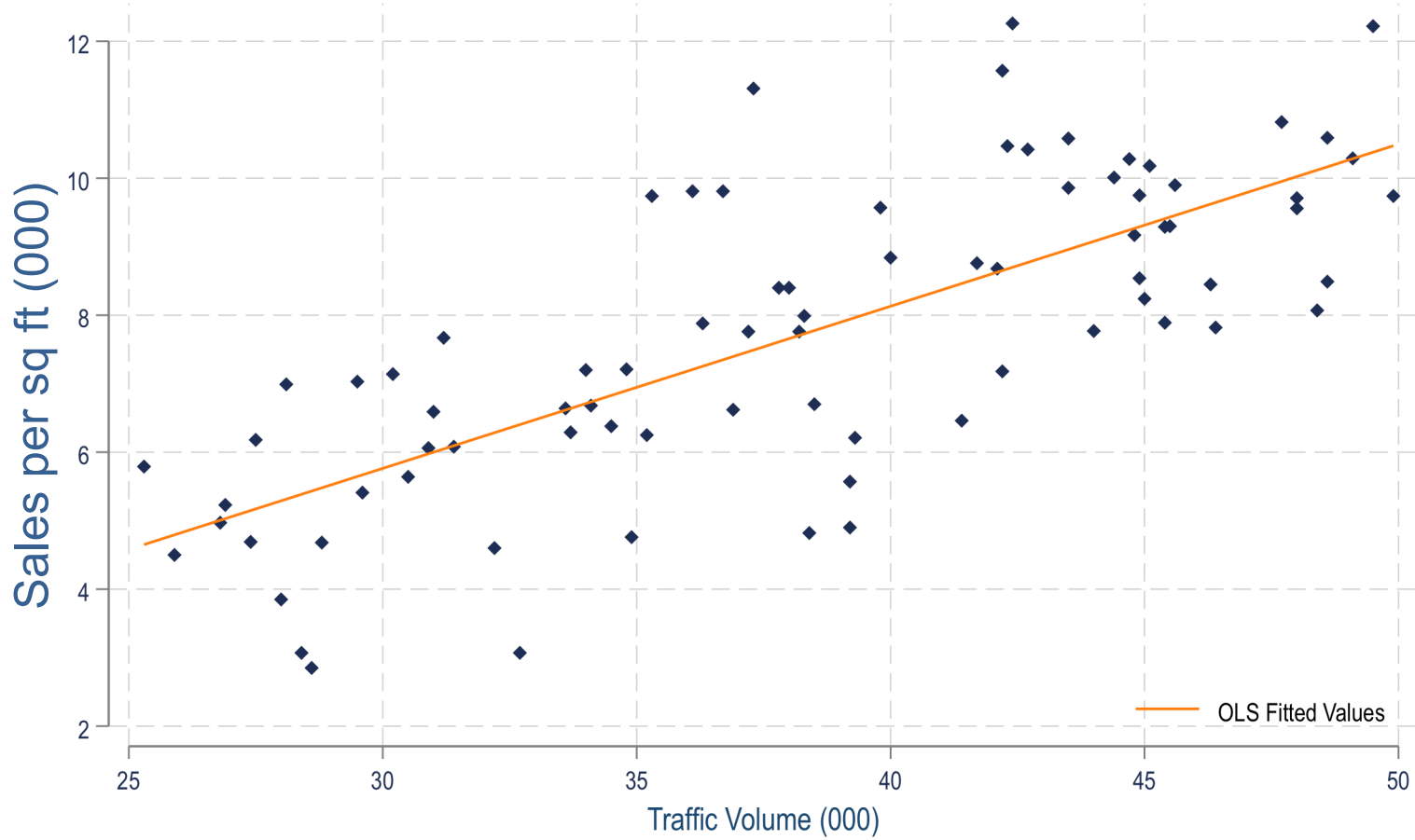
Locating a New UNIQLO Store

Scatterplot



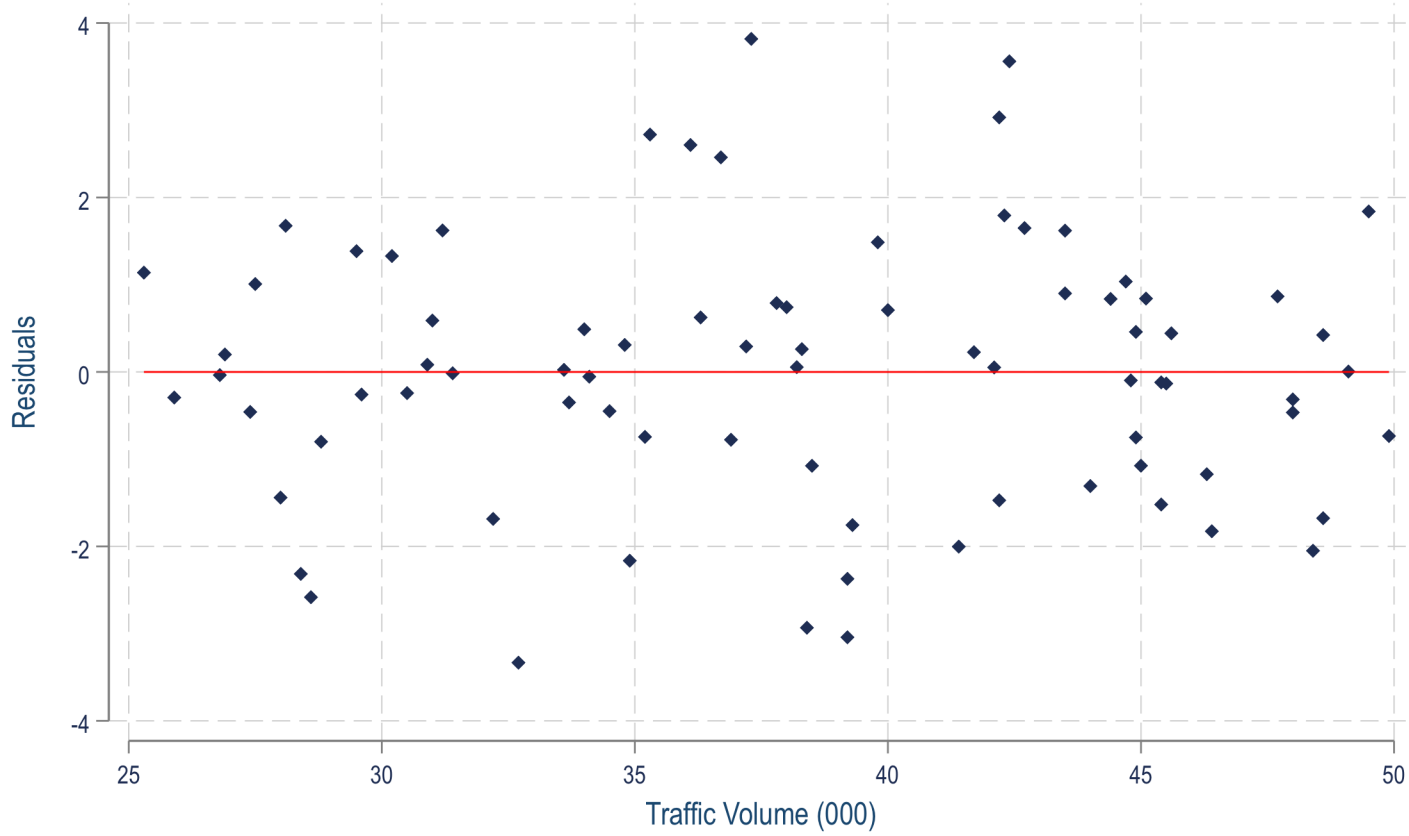
Locating a New UNIQLO Store

Regression Line



Locating a New UNIQLO Store

Residual Plot



Locating a New UNIQLO Store

Regression Output

R Square	0.55
Adjusted R Square	0.54
Standard Error	1.51
Observations	80

	df	*SS*	*MS*	*F*	*p(F)*	
Regression	1	214.83	214.83	94.79	0.00	
Residual	78	176.77	2.27			
Total	79	391.59				
	b	***s.e.***	***t***	***p***	***95% LB***	***95% UB***
Intercept	-1.34	0.95	-1.41	0.16	-3.22	0.54
Traffic	0.24	0.02	9.74	0.00	0.19	0.29

Inference in regression: hypothesis tests

Standard hypothesis tests from regression software:

Slope: H_0: $\beta_1 = 0$, use $t = \dfrac{b_1}{se(b_1)}$

Intercept: H_0: $\beta_0 = 0$, use $t = \dfrac{b_0}{se(b_0)}$

The t distribution in this case has n-2 degrees of freedom

Why is software automatically testing this hypothesis?

Inference in regression: UNIQLO data

- The t-statistic for the slope is, $t = \frac{b_1}{se(b_1)} = \frac{0.24}{0.02} = 9.74$
- With p-value of < 0.0001 indicates that the slope is significantly different from zero
- The t-statistic for the intercept is, $t = \frac{b_0}{se(b_0)} = \frac{-1.34}{0.95} = -1.41$
- With p-value of 0.16 indicates that the intercept is not significantly different from zero
- NOTE: These are two-sided tests!

Inference in Regression

Confidence Intervals

The 95% confidence interval for β_1 is:

$$b_1 \pm t_{0.025,n-2} * se(b_1).$$

The 95% confidence interval for β_0 is:

$$b_0 \pm t_{0.025,n-2} * se(b_0).$$

Locating a New UNIQLO Store

Confidence Intervals

	b	*s.e.*	*t*	*p*	*95% LB*	*95% UB*
Intercept	-1.34	0.95	-1.41	0.16	-3.22	0.54
Traffic	0.24	0.02	9.74	0.00	0.19	0.29

In the UNIQLO example, the 95% confidence interval for β_1 is…

$$b_1 \pm t_{0.025,n-2} * se(b_1)$$

$$0.2367 \pm 1.992 * .0243$$

$$[0.188, 0.285]$$

Inference in regression: prediction intervals

Back to the Original Question

What would we predict for UNIQLO sales at a location with 50,000 cars driving by per day?

Our best guess is $10,660 per day.
Why? Because -1.34 + (0.24) (50) =10.66.

But could sales be as low as 9,000? As low as 5,000?

- To answers questions like this we need a prediction interval

Inference in regression: prediction intervals

A **prediction interval** is an interval designed to hold a fraction (usually 95%) of the values of the response for a given value of x.

A prediction interval *differs* from a confidence interval because it makes a statement about the location of a new observation rather than a population parameter.

Prediction Intervals

- A simple approximation for a 95% prediction interval is,

$$\hat{y} \pm 2\, s_e.$$

- Prediction intervals are reliable within the range of observed data – don't extrapolate too much!

- They are also sensitive to the assumptions of constant variance (see more "correct" s_e formula in next slides)

Locating a Franchise Outlet

Prediction Interval

What is the 95% prediction interval for the amount of sales for a store with 50,000 daily cars driving by?

From the regression output we know that s_e = 1.51.

Thus the 95% prediction interval is 10.66 ± (2)(1.51).

Or, approximately 7,600 to 13,700 dollars per day.

Prediction Interval Formula

- Typically we will rely on approximate formula

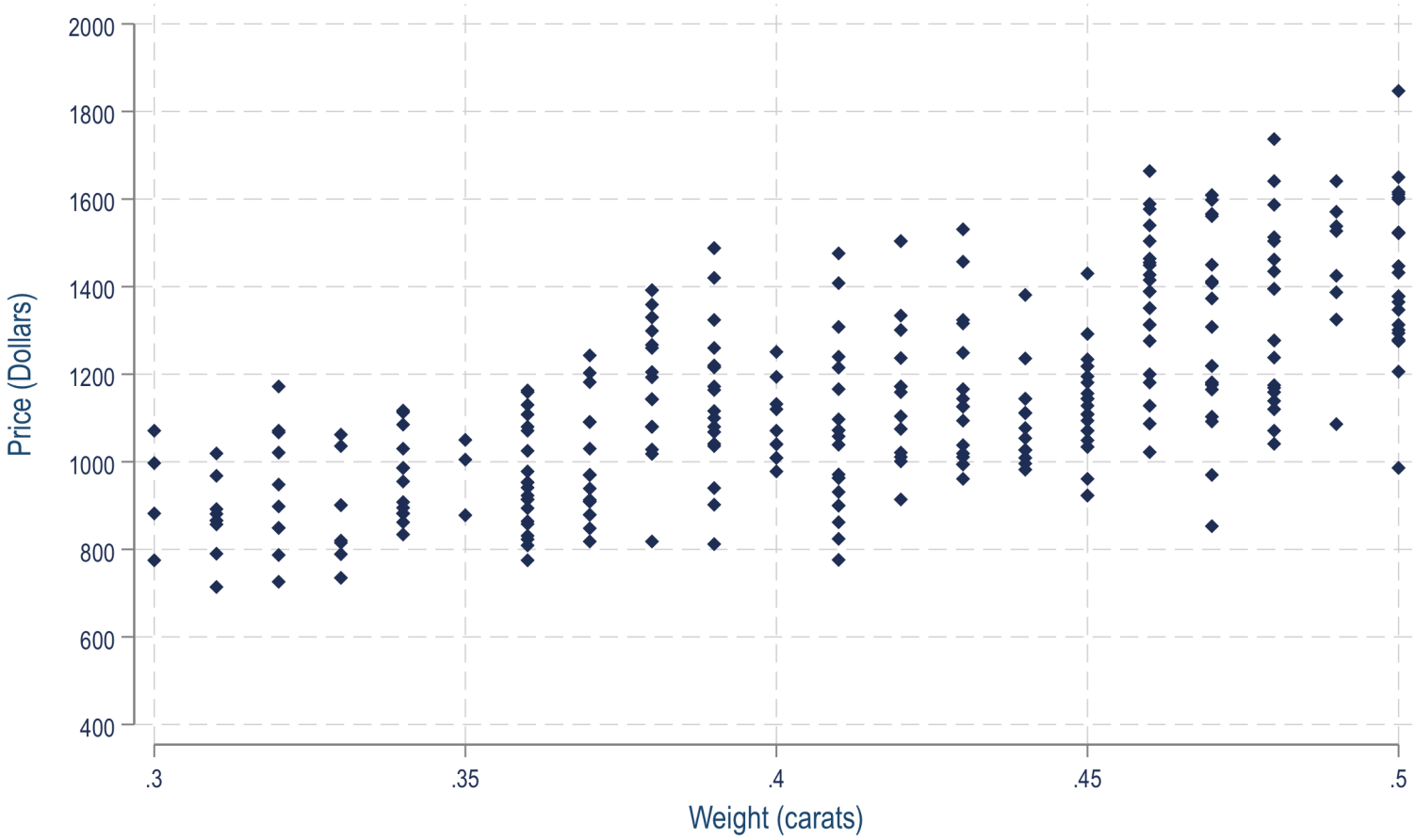
$$\hat{y} \pm 2\, s_e.$$

- True formula for standard error of prediction relies on a more complicated formula (that you should "know about" rather than "know"):

$$se(\hat{y}_{new}) \approx s_e \sqrt{1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{(n-1)s_x^2}} \approx s_e$$

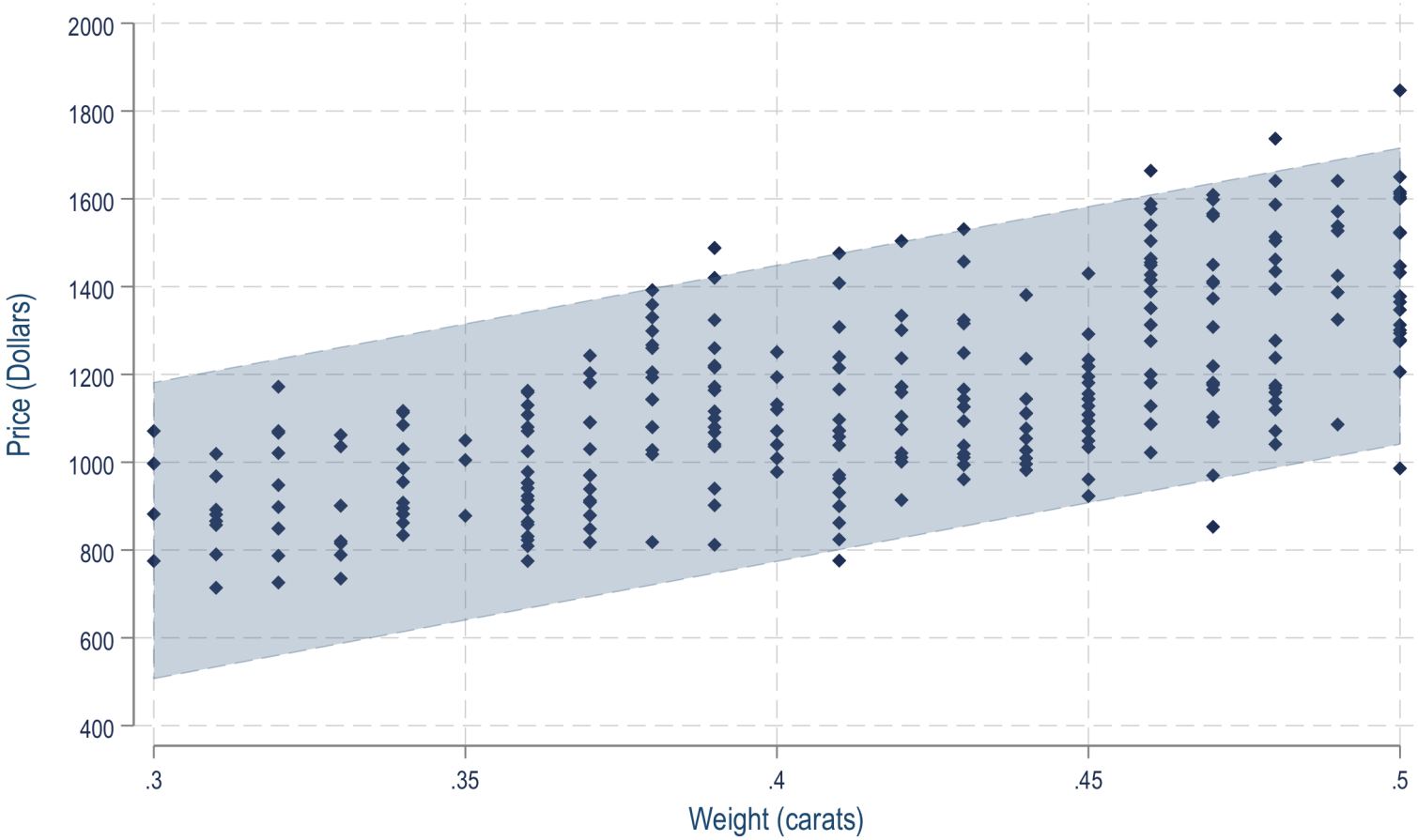
Prediction intervals: extrapolation

Back to diamonds



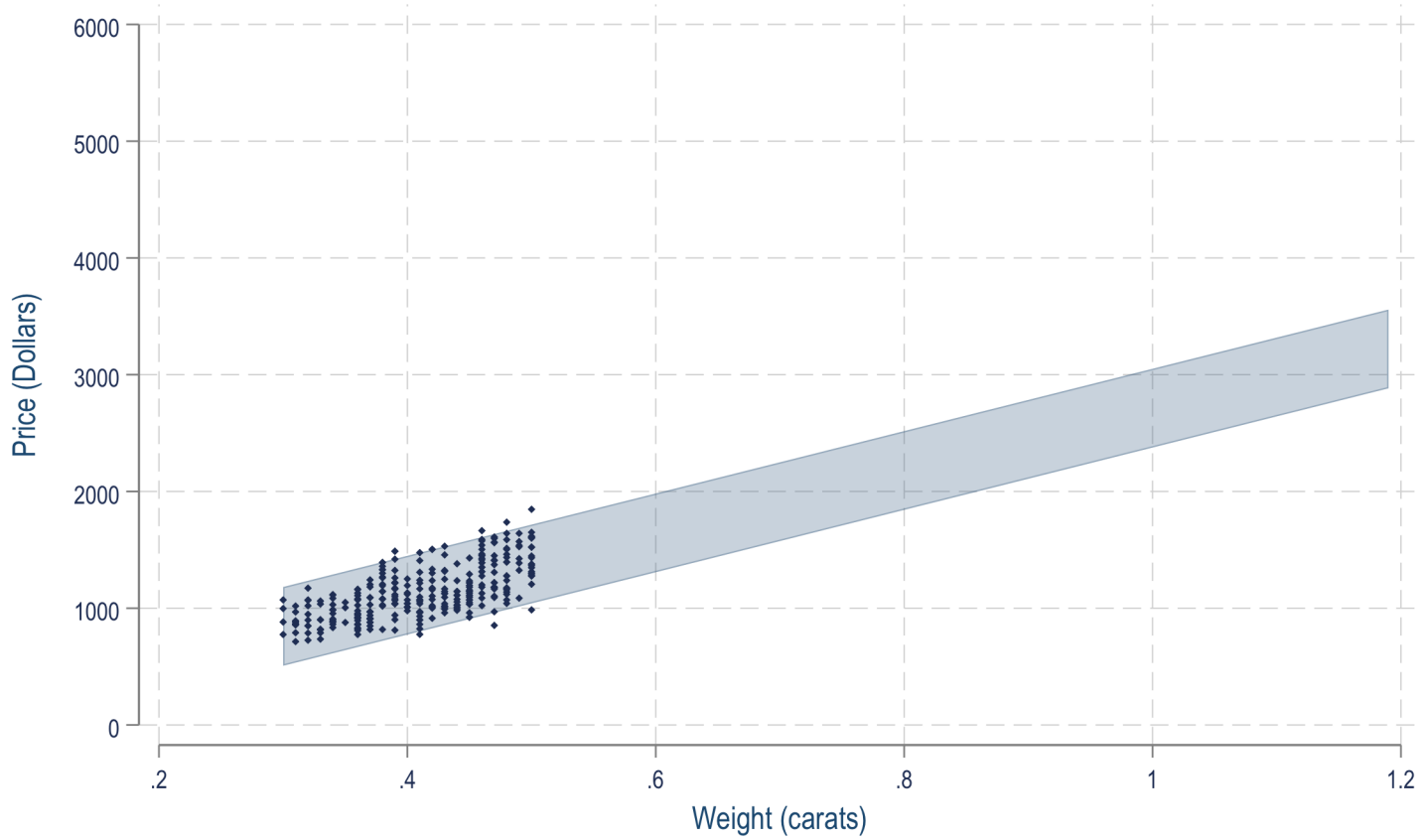
Prediction intervals: extrapolation

Back to diamonds



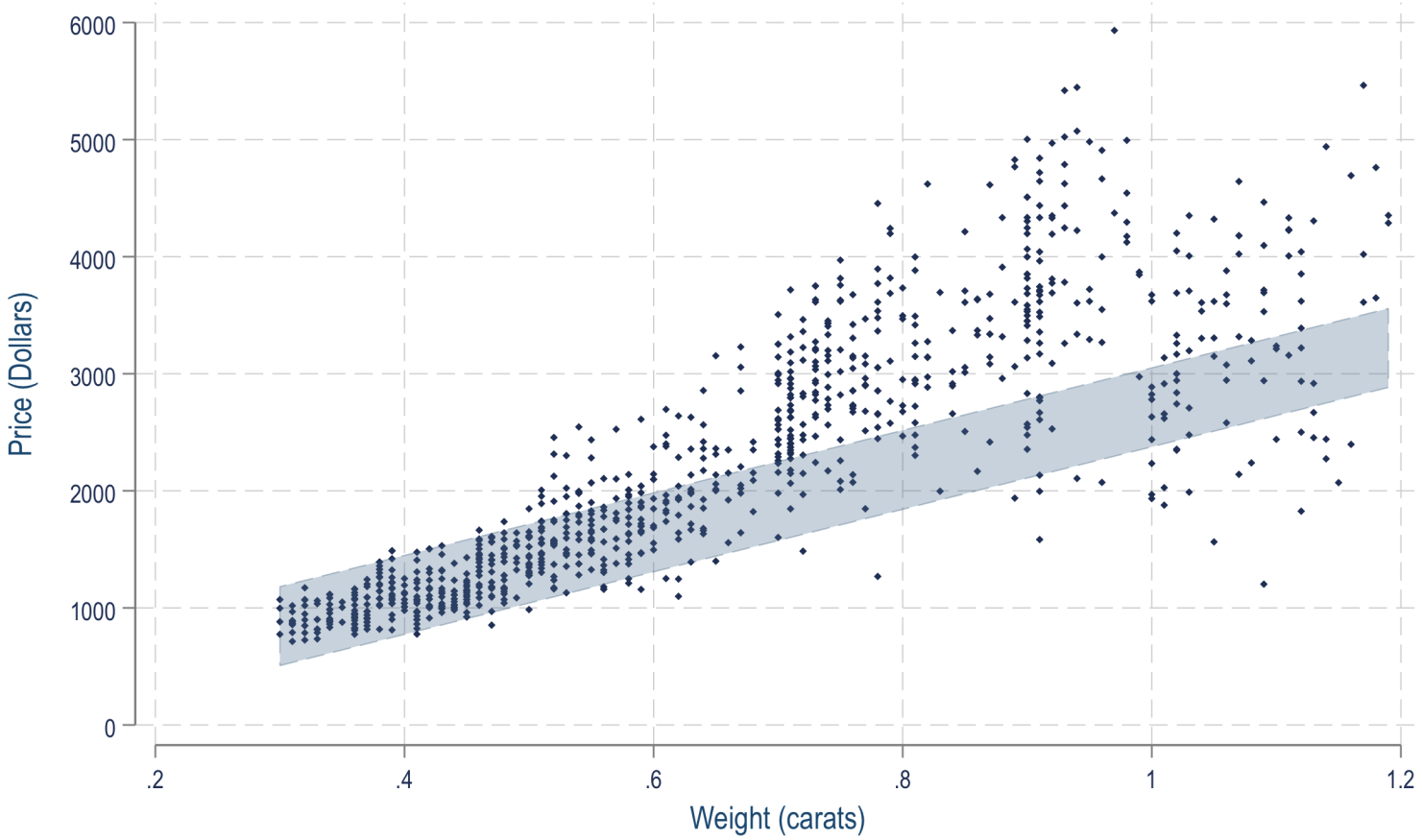
Prediction intervals: extrapolation

Back to diamonds



Prediction intervals: extrapolation

Back to diamonds



Best Practices

- But also know that modest deviations from these assumptions are ok.
- Rely on software to calculate standard errors for regression estimates
  - Most analysts use “robust” standard errors in regression models
- Use prediction intervals to predict y for particular observations.
- Be careful when extrapolating.

Big Picture / Recap

Started by thinking about inference based on limited information.

- Is your production process working as it should?
- Initially, the inference problem was very simple – we had some benchmark given to us and wondered whether what we observed in a sample was consistent with the benchmark.

Then, we made things more difficult:

- What fraction of your customers are willing to pay for a new feature on your app?
- Try to make inferences about something we didn't observe, based on limited information we did observe.

Next, we made things even more difficult:

- Testing differences between two groups, neither of which we observed completely.

Now, we want to make inferences about more complex relationships:

- What linear model exists in reality?
- Well, we don't know. We can collect some data, and then we'll make inferences about unknown parameters and we'll run statistical tests on these unknown parameters.

The point is that to understand what we are doing with regressions, you need to build up from simple inferences to complex ones.

This class: how to think about regression variation – again, you have limited information so you are making an *inference* about a population relationship from a sample